

Multiple Choice

1. **Answer:** D

Options: A) The wait is expected to be short.; B) A busy-wait loop is easier to code than an interrupt handler.; C) There is no other work for the processor to do.; D) The program executes on a time-sharing system.

Note: Correct option: D - The program executes on a time-sharing system.

2. **Answer:** D

Options: A) All bit strings whose number of zeros is a multiple of five; B) All bit strings starting with a zero and ending with a one; C) All bit strings with an even number of zeros; D) All bit strings with more ones than zeros

Note: Correct option: D - All bit strings with more ones than zeros

3. **Answer:** D

Options: A) Session IDs; B) Registry keys; C) Network communications; D) SQL queries based on user input

Note: Correct option: D - SQL queries based on user input

4. **Answer:** B

Options: A) No, I cannot compute the key.; B) Yes, the key is $k = m \text{ xor } c$.; C) I can only compute half the bits of the key.; D) Yes, the key is $k = m \text{ xor } m$.

Note: Correct option: B - Yes, the key is $k = m \text{ xor } c$.

5. **Answer:** B

Options: A) front-door; B) backdoor; C) clickjacking; D) key-logging

Note: Correct option: B - backdoor

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It generates each different input by modifying a prior input'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.5'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'WPS'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Integrity'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Network or transport layer'.

Long Form Response

11. **Answer:** `def is_Perfect_Square(n) : | return True | return False`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def min_Operations(A,B): | return B - 1`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key